Cyber Security Threat Analysis

Using Microsoft Power BI

Project Overview

Cybersecurity incidents can have significant financial, operational, and data-related consequences for organizations.

In this project, I analyzed cybersecurity incident data to understand:

- Trends in cybersecurity incidents over time
- Major threat types affecting the organization
- Distribution of threat severity
- Financial and data breach impact
- Assets and departments affected by incidents
- Incident resolution performance

Using **Microsoft Power BI**, I transformed the raw dataset into an interactive **three-page business intelligence solution**.

Problem Statement

Organizations generate large amounts of cybersecurity incident data, but raw data alone does not provide an easy way to identify important patterns.

The key challenge is to understand:

- Which threats occur most frequently?
- Which threats contribute most to financial impact?
- How severe are the incidents?
- Which assets are most affected?
- Which departments experience the most incidents?
- How long does it take to resolve incidents?

This project aims to transform raw cybersecurity data into an **interactive analytical solution** that can support security monitoring and data-driven decision-making.

Project Objectives

The main objectives of this project are:

- Analyze cybersecurity incident trends over time
- Identify major threat types
- Analyze incidents by severity category
- Measure financial impact and data breached
- Analyze incidents across different asset types
- Examine departmental incident distribution
- Evaluate incident resolution performance
- Build an interactive Power BI dashboard
- Demonstrate practical Power BI, DAX, Power Query and data modeling skills

Dataset Description

- The project uses a **Kaggle cybersecurity incident dataset** containing information related to security incidents, threats, assets, employees and their impact.
- The dataset contains approximately **5,000 incident records**.
- **Major data areas include:**
- **Incident Information**
- Incident ID, Detection Date & Time, Incident Status, Financial Impact, Data Breached, Time to Resolve
- **Threat Information**
- Threat Type, Threat ID, Severity, Vulnerability information
- **Asset Information**
- Asset ID, Asset Type, Vulnerability information
- **Employee / Organization Information**
- Employee ID, Department, Security Training information
- **Source:** Kaggle
- *Replace "Kaggle cybersecurity incident dataset" with the exact Kaggle dataset name in your final PPT.*

Data Understanding & Quality Check

Before building the dashboard, I performed an initial data quality assessment.

Checks performed:

- Reviewed dataset structure and columns
- Checked for missing/null values
- Verified data types
- Checked for duplicate records
- Reviewed categorical values
- Examined relationships between tables
- Identified columns useful for business analysis
- **Data Quality Result**
- **5,000 incident records retained**
- No null values requiring removal
- No incorrect data types identified
- Duplicate validation performed using Incident ID
- The data was suitable for further transformation and modeling.

Data Cleaning & Preparation

Several Power Query transformations were performed to prepare the dataset for analysis.

Cleaning

- Renamed columns using readable business-friendly names
- Verified and standardized data types
- Checked and handled duplicate records
- Validated categorical values

Feature Creation

- Created additional analytical columns:
- **Detection Date**
- **Detection Time**
- **Threat Severity Category**
- **Vulnerability Score Category**

Objective

- The goal was to keep the dataset clean while creating only the additional fields required for meaningful dashboard analysis.

Power Query Transformations

Power Query Workflow

The data preparation process followed this workflow:

- **Import → Inspect → Clean → Transform → Validate → Load**

Power Query was used for:

- Column renaming
- Data type validation
- Duplicate checking
- Date & time separation
- Conditional/category creation
- Data quality validation

Data Modeling

Star Schema

The Power BI model was designed using a **Star Schema**.

Fact Table

- **Fact_Incidents**
- Contains incident-level transactional data such as:
- Financial Impact
- Data Breached
- Resolution Time
- Detection Date
- Detection Time

Dimension Tables

- **Dim_Threats**
- **Dim_Assets**
- **Dim_Employees**
- **Dim_Date**

Relationships

- The dimension tables connect to the central fact table using **one-to-many relationships** with single-direction filtering.

Date Dimension

Date Dimension & Time Analysis

A dedicated Dim_Date table was created using DAX.

The Date Dimension contains:

- Date, Year, Quarter, Month Number, Month Name, Month-Year, Week Number, Day, Day Name, Day of Week,

  Weekend indicator

Benefits

- Enables consistent time-based analysis
- Supports monthly and yearly reporting
- Avoids repeating date attributes in the fact table
- Improves model organization
- Supports Power BI time intelligence
- The Date table was also **marked as a Date Table** and connected to the incident fact table.

DAX Measures & KPI Development

DAX Measures

Reusable DAX measures were created instead of relying only on visual-level calculations.

Key Measures

- Total Incidents
- High Severity Incidents
- Total Financial Impact
- Total Data Breached
- Average Resolution Time
- Total Assets
- Total Employees
- Total Threat Types
- Average Financial Impact per Incident
- Average Data Breached per Incident

Dashboard Design & Navigation

Interactive Power BI Solution

- The final report contains a **Home Page + 3 analytical dashboards**.

Home Page

- Provides navigation to:
- Executive Overview
- Threat Analysis
- Asset & Employee Analysis

Interactive Features

- Page navigation buttons
- Slicers
- Cross-filtering
- Interactive visuals
- KPI cards
- Conditional formatting
- Multiple chart types

The report was designed with a consistent **cybersecurity-themed visual style**.

Dashboard 1: Executive Overview

Executive Overview

- The first dashboard provides a high-level view of the cybersecurity environment.

Main Components

- Total Incidents KPI
- High Severity Incidents KPI
- Financial Impact KPI
- Data Breached KPI
- Incident Trend Over Time
- Incidents by Threat Type
- Incident Status Distribution
- Incidents by Department

Filters

- Year
- Severity Category

Purpose

- Provides an executive-level summary of incident volume, impact and distribution.

Dashboard 2: Threat Analysis

Threat Analysis

- The second dashboard focuses specifically on threat patterns and their impact.

Main Components

- Total Threat Types
- High Severity Incidents
- Average Financial Impact
- Average Data Breached
- Threat Ranking Over Time
- Financial Impact by Threat Type
- Threat Severity Analysis
- Financial Impact Breakdown

Power BI Visuals Used

- **Ribbon Chart**
- **Waterfall Chart**
- **Matrix**
- **Decomposition Tree**

Filters

- Threat Type
- Severity Category

Purpose

- Enables deeper analysis of threat frequency, severity and financial impact.

Dashboard 3: Asset & Employee Analysis

The third dashboard focuses on the organizational and asset-level distribution of cybersecurity incidents.

Main Components

- Total Assets
- Total Employees
- Total Incidents
- Average Resolution Time
- Incidents by Asset Type
- Assets by Vulnerability Risk
- Incident Status
- Average Incident Resolution Time

Visuals Used

- Bar Chart
- Pie/Donut-style visualization
- Funnel Chart
- Gauge Chart

Filters

- Department
- Asset Type

Purpose

- Helps identify affected asset types, vulnerability distribution and incident resolution performance.

Key Business Insights

Key Insights

- The dashboard analysis revealed several important patterns.

Incident Trends

- Monthly incident volume fluctuated throughout the year.
- **November recorded the highest incident volume at 443 incidents.**
- **February recorded the lowest at 363 incidents.**

Threats

- **Social Engineering** represents one of the largest contributors to total incidents.
- **Malicious Code** is also a major contributor to incident volume and financial impact.

Severity

- **1,259 incidents** were categorized as High/Critical based on the project's severity measure.
- Low and Medium vulnerability categories represent a larger share of assets than High vulnerability.

Impact

- Total financial impact is approximately **1.82M K**.
- Total data breached is approximately **73.8K GB**.

Resolution

- Average incident resolution time is approximately **120.4 hours**.

Who Can Use This Dashboard?

Potential Stakeholders

This dashboard can be useful for:

- 🛡️ **Security Teams**
- Monitor incident patterns, severity and threat types.
- 👨‍💼 **Management**
- Understand financial and operational impact.
- 💻 **IT / Infrastructure Teams**
- Identify affected asset types and vulnerabilities.
- 👥 **HR / Training Teams**
- Explore departmental and employee-related security patterns.
- 📊 **Business Analysts**
- Perform exploratory analysis and identify trends.

Business Impact & Future Improvements

Business Impact

- The dashboard enables stakeholders to:
- Monitor cybersecurity incidents from a single reporting interface
- Identify major threat categories
- Understand financial and data exposure
- Compare affected departments and assets
- Analyze incident resolution performance
- Perform interactive exploration using filters

Future Improvements

- The project could be extended by adding:
- Predictive threat analysis
- Machine Learning-based risk prediction
- Integration with SIEM/security monitoring systems
- Incident-level drill-through analysis

Conclusion

This project demonstrates an end-to-end **Power BI data analytics workflow**:

- **Raw Data**
  ↓
  Data Cleaning
  ↓
  Power Query Transformation
  ↓
  Star Schema Modeling
  ↓
  DAX Measures & KPIs
  ↓
  Interactive Dashboards
  ↓
  Business Insights

Conclusion

Through this project, I analyzed cybersecurity incidents from multiple perspectives including **threats, severity, financial impact, data exposure, assets, departments and resolution performance**.

The project helped me strengthen my practical understanding of:

- Power Query
- Data Modeling
- DAX
- Data Visualization
- Dashboard Design
- Interactive Reporting
- Business Intelligence Storytelling